

SUMMARY TABLE: BIOLOGY GRADE 10

Sub-Strand 1.3: Cell Biology — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 10	
Sub-Strand	Sub-Strand 1.3: Cell Biology
Driving Question	DRIVING QUESTION: "Why do the cells in my body — and in the sukuma wiki on my plate — look so different from each other, and how does each cell's structure allow it to do its specific job?"

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 What Is a Cell? Launching the Phenomenon — Blood, Sukuma Wiki, and the Mystery of Life's Building Blocks	Students observed two microscope images side by side: a stained human blood smear showing biconcave anucleate red blood cells and nucleated white blood cells, and a sukuma wiki (kale) leaf cell showing a rectangular shape, chloroplasts, a large central vacuole, and a visible cell wall. Students completed a structured observation table comparing features across three cell types.	Students learned that cells can differ dramatically in shape, size, and internal structures while still being classified as cells. They were introduced to the historical development of cell theory through Hooke (1665), Schleiden (1838), Schwann (1839), and Virchow (1855), and practised distinguishing between scientific observations and the conclusions drawn from them.	Students used Claim—Evidence—Reasoning (CER) to argue — based only on today's image evidence — which structures they believe are universal to all cells. They revised their initial 'generic cell' sketch using a different colour to show evidence-based changes, recorded a Model Changelog, and posted questions and revised predictions to the newly created Driving Question Board (DQB).
Lesson 2 Seeing Cells for the First Time: Microscopy Practical 1	Onion epidermal cells are rectangular, have a thick outer wall, and show a visible vacuole when stained with iodine. Cheek cells are irregular and rounder in shape, do not have a thick outer wall, and show a faint nucleus when stained with methylene blue. The onion cell did NOT show the green colour seen in the sukuma wiki leaf cell from Lesson 1.	Both plant and animal cells share a cell membrane and cytoplasm. Plant cells (onion) also have a cell wall that gives them a rigid rectangular shape, and a large vacuole. Animal cells (cheek) lack a cell wall and have a more irregular shape. Structures visible under a light microscope include the cell wall, nucleus, and vacuole — but many internal structures are too small to see at this magnification.	The rectangular shape of the onion cell is explained by the presence of a rigid cell wall — a structure the cheek cell does not have. The absence of green colour in the onion cell (even though it is a plant cell) raises a new question that will be explained in Lesson 4 when students study chloroplasts and where in a plant they are found.
Lesson 3 What Is Inside the Cell Doing All the Work? — Organelle Structure &	Learners examined annotated electron micrograph images and close-up TEM images of individual animal cell organelles (nucleus, mitochondria, ribosomes, rough	Learners identified and explained the structure and function of 10 animal cell organelles, mapping each to a department in a Nairobi Industrial Park factory analogy.	Learners explained that each organelle's structure is directly related to its function — for example, the folded inner membranes (cristae) of the mitochondria maximise

Function (Animal Cell)	and smooth ER, Golgi apparatus, lysosomes, centrioles), recording structural features they could directly see in an Organelle Evidence Table before receiving any expository function information.	Learners traced the protein synthesis functional pathway (nucleus → ribosome → rough ER → Golgi → cell membrane) and connected organelle knowledge back to the anchoring phenomenon by explaining why the Kisumu blood smear red blood cell lacks a nucleus and what functional consequences follow.	surface area for ATP production, just as a generator room is sized to power the whole factory. They explained the anchoring phenomenon observation — the enucleated red blood cell — as a specialisation trade-off: losing the nucleus (Managing Director's office) sacrifices the ability to make proteins or replicate, but maximises internal space for haemoglobin and oxygen transport, a function critical to its role in the bloodstream.
Lesson 4 Organelle Structure & Function — The Plant Cell: How Is a Sukuma Wiki Cell Like and Unlike a Human Cell?	Students observed two bean seedlings — one turgid and upright (well-watered, like a healthy shamba in Kericho) and one wilted and drooping (water-deprived, like a farm during a dry spell in Machakos) — and connected the visible wilting to changes in the central vacuole inside plant cells. Students also revisited their own Lesson 3 microscope observation drawings of sukuma wiki leaf cells, identifying three structures absent in the blood smear from the anchoring phenomenon.	Students learned that plant cells contain three organelles NOT found in animal cells: (1) Chloroplasts — green, oval organelles containing chlorophyll that capture sunlight to make food via photosynthesis, explaining why sukuma wiki is green and can produce its own glucose while blood cells cannot; (2) Cell Wall — a rigid outer layer made of cellulose outside the cell membrane, explaining why sukuma wiki leaf cells appear rectangular and have a defined shape, unlike the flexible, round red blood cells in the blood smear; (3) Central Vacuole — a large, water-filled sac that creates turgor pressure to keep plant cells firm and upright, explaining why a bean plant wilts when water is scarce and why the large central sac observed in Lesson 3 was pushing organelles to the edges of the sukuma wiki cell.	Students can now partially explain the driving question phenomenon: the sukuma wiki leaf cell looks rectangular, green, and has a large central sac because it contains a cell wall (cellulose rigidity), chloroplasts (photosynthesis / green pigment), and a central vacuole (turgor pressure / water storage) — all of which are absent in the red blood cells and white blood cells seen in the Kisumu County Hospital blood smear. Despite these dramatic structural differences, both cell types share a cell membrane, cytoplasm, and nucleus (in most cases), confirming that they are both cells — but specialised for completely different jobs. Students have produced Version 1 of their cumulative 2D comparative cell diagrams, which they will continue to develop across Lessons 5–8.
Lesson 5 Are Bacteria Cells the Same as Our Cells? Prokaryotes vs. Eukaryotes	Students examined labelled diagrams of E. coli (prokaryote) and a human cheek cell (eukaryote) on the ARES evidence sheet, and completed a structured 10-feature comparison table identifying features present in both, only in prokaryotes, or only in eukaryotes.	Prokaryotic cells (e.g., bacteria like Lactobacillus in fermenting uji) have no membrane-bound nucleus and no membrane-bound organelles; their DNA sits free in the cytoplasm in a region called the nucleoid. Eukaryotic cells (e.g., human cheek cells, sukuma wiki leaf cells) have a membrane-bound nucleus and specialised organelles. Despite these differences, ALL cells share four universal features: a cell membrane, cytoplasm, DNA, and ribosomes.	Using a Claim–Evidence–Reasoning paragraph, students explained that bacteria ARE real cells — despite lacking a nucleus — because they possess all four universal cell features. Students connected this to the anchoring phenomenon by recognising that the bacteria fermenting uji are prokaryotes whose structural simplicity (no nucleus, no membrane-bound organelles) is fundamentally different from the nuclear loss seen in specialised eukaryotic RBCs, and updated their cumulative cell models to include a prokaryote column alongside the generic eukaryotic, specialised animal, and specialised plant cell columns.

Lesson 6 Cell Specialisation — Why Don't All Cells Look the Same?	Students examined six Evidence Cards, each containing a labelled diagram, a Kenya-relevant context clue, and a partial function statement for one specialised cell (red blood cell, guard cell, root hair cell, muscle cell, nerve cell, sperm cell). They recorded structural observations in a two-column STRUCTURE POSSIBLE FUNCTION table and participated in a CER share-out where each group presented a Claim–Evidence–Reasoning explanation for one specialised cell.	Students learned that specialised cells deviate from the generic cell model in specific, purposeful ways: red blood cells lack a nucleus to maximise haemoglobin space for oxygen transport; guard cells are bean-shaped to open and close stomatal pores in sukuma wiki leaves; root hair cells are elongated to increase surface area for water absorption in bean seedlings; muscle cells are elongated with many mitochondria to power contraction; nerve cells have long axons to transmit signals rapidly across the body; and sperm cells have a flagellum and mitochondria-rich midpiece to power motility toward an egg. Students also applied structure-function reasoning to the sickle cell supporting phenomenon, explaining that the crescent shape causes cells to stack and block capillaries, reducing oxygen delivery to tissues.	Students constructed written CER (Claim–Evidence–Reasoning) explanations linking specific structural features of specialised cells to their functions, revised their cumulative cell models by adding a colour-coded Specialisation Layer with divergence arrows, and updated the Driving Question Board with new 'I can now explain' and 'I still wonder' statements. Through these activities, students built a complete explanatory answer to the driving question: cells look different because their structures are precisely shaped to perform their specific jobs, and any disruption to that structure — as in sickle cell anaemia — disrupts function in measurable, life-affecting ways.
Lesson 7 Building Our Complete Cell Model: Synthesising Evidence to Explain the Anchoring Phenomenon	Students observed their group's material kit (avocado seeds, bottle tops, plastic bags, maize grains, green paper, dried beans, cardboard, etc.) alongside a reference card showing labelled plant and animal cell diagrams. They observed other groups' completed 3D models during structured presentations and peer assessment. They observed the full Driving Question Board from Lessons 1–7 during the DQB Tripartite Sort.	Students learned how to synthesise structural and functional knowledge of all major organelles into a single coherent 3D model. They learned that plant cells uniquely contain cell walls, chloroplasts, and a large central vacuole, while animal cells lack these but may contain centrioles. They learned that specialised cells (red blood cells, white blood cells, sukuma wiki leaf cells) are DEVIATIONS from a generic cell model, with specific structures added or removed to optimise a particular function. They learned to distinguish 'answered' from 'partially answered' from 'bridge' questions — understanding that unanswered questions are the starting point for future scientific inquiry.	Students explained why the red blood cell in the Kisumu hospital blood smear has no nucleus (loss of nucleus maximises haemoglobin space for oxygen transport — a specialisation away from the generic cell model). They explained why sukuma wiki leaf cells are rectangular, have thick green chloroplasts, and a large central vacuole (plant cell features: cell wall for rigidity and support, chloroplasts for photosynthesis, vacuole for turgor pressure and water storage). They explained what ALL cells in the phenomenon share: a cell membrane, cytoplasm, and genetic material — the irreducible minimum of a 'cell'. They explained how their 3D model, built from Kenyan everyday materials, represents a 'generic' cell that is the reference point from which all specialised cells deviate.
Lesson 8 Driving Question Synthesis: So Why DO Cells Look So Different?	We revisited the anchoring phenomenon — the blood smear showing red blood cells without nuclei next to white blood cells with large lobed nuclei, and the sukuma wiki leaf cell with its rectangular shape, chloroplasts, and large central vacuole. We observed a three-cell comparison card identifying	All cells share core structures including a cell membrane, cytoplasm, ribosomes, and DNA, because these are essential to the basic processes of life including protein synthesis, energy use, and reproduction. Plant cells are distinguished by a cell wall made of cellulose, chloroplasts for	We can now answer the driving question with evidence: the cells in our body and in the sukuma wiki look different from each other because specialisation requires each cell to prioritise certain functions over others, and the most efficient way to do that is to modify or remove structures that are

	shared structures, type-specific structures, and specialisation modifications for each cell. We observed the completed Driving Question Board from all eight lessons and reviewed which questions had been answered and which remain as bridges to Sub-Strand 1.4.	photosynthesis, and a large central vacuole for turgor pressure — all visible in the sukuma wiki leaf. Animal cells may possess centrioles for cell division. Specialised cells are not defective generic cells — they are optimised deviations: the red blood cell loses its nucleus and gains haemoglobin capacity; the guard cell changes shape to control the stomatal opening in the sukuma wiki leaf; the root hair cell extends to maximise absorption surface area. Sickle cell disease demonstrates that even a small structural change — caused by a single gene mutation — completely disrupts a specialised cell function and has significant health consequences in Kenyan communities where the sickle cell trait is present.	not needed while amplifying or adding structures that are. The red blood cell without a nucleus is not missing something — it has made space for more haemoglobin so it can carry more oxygen. The sukuma wiki leaf cell is not more complex than a cheek cell — it has additional structures because it must harvest light energy from the sun, something an animal cell never needs to do. All of these cells — red blood cell, white blood cell, leaf cell, root hair cell — are still cells because they all share the fundamental architecture of membrane, cytoplasm, and genetic material. The difference is in what has been added, removed, or modified in service of a specific job in a specific organism living in a specific environment.
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